

# The Terminated Lossless Transmission Line

## Problem 2: The Short Circuit Terminated Ideal Transmission Line

### Problem Statement

Show graphically in ADS the changes in reactance, magnitude voltage and magnitude current with respect to line length, for an ideal transmission line that is terminated in a short circuit.

### Solution

#### Strategy

Use the ideal transmission model within ADS and connect to a 2-volt voltage source with 50-Ohm generator impedance. Terminate the line in a short circuit, and use node voltages to find the changes in reactance, voltage and current along the transmission line with respect to electric length.

#### What to expect

The impedance mismatch between the 50-Ohm transmission line and short circuit termination will cause a full reflection. The resulting impedance seen looking into the transmission line will vary with its length. Using the characteristic equation for input impedance for a transmission line, the results can be examined before simulation.

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} \quad \text{Equation 1.9}$$

For the short circuited transmission line,  $Z_L = 0$ , and the equation reduces to  $Z_{in} = jZ_0 \tan \beta l$ . At the frequency where the line is an integer multiple of  $\lambda$ ,  $\beta l = \frac{2\pi}{\lambda} \cdot n\lambda = 2n\pi$ , making  $\tan \beta l = 0$ . As a result, we expect to see the short circuit termination only,  $Z_{in} = 0$ . The voltage and current will measure 0 V and 40 mA (2 volts/50 Ohm), and the reactance will be 0 Ohms. At the frequency where the line is an odd multiple of a quarter of a wavelength,  $\beta l = \frac{2\pi}{\lambda} \cdot n\frac{\lambda}{4} = n\frac{\pi}{2}$ ;  $n = \text{odd}$ , making  $\tan \beta l \rightarrow \infty$ . As a result, the reactance will be infinite, making the line look like an open circuit. The incident signal from the source and reflected signal from the termination add constructively, resulting in a peak in the standing wave. The voltage and current will measure 2 volts (1 volt + 1 volt) and 0 mA (40 mA + -40mA). When the line length is  $0.5\lambda$ , the transmission line looks like a short again,  $Z_{in} = 0$ . There will be a phase change, but because we are plotting the magnitude voltage and current, the values will be the same as the  $0-\lambda$  line length.

#### Execution

Set up the schematic for the short circuit transmission line with an AC voltage source and voltage probes between the generator and transmission line (Figure 1-18). The voltage probes are found in the Probe Components palette. Rename the probes something more descriptive than V\_Probe1/2, like V\_source and V\_inc. The node can also be named to designate the voltage source, but for this analysis, the probe provides a better physical realization of the measurement.

The voltage source should send out 2 volts to keep the voltage division simple between a 50-Ohm generator and 50-Ohm line.

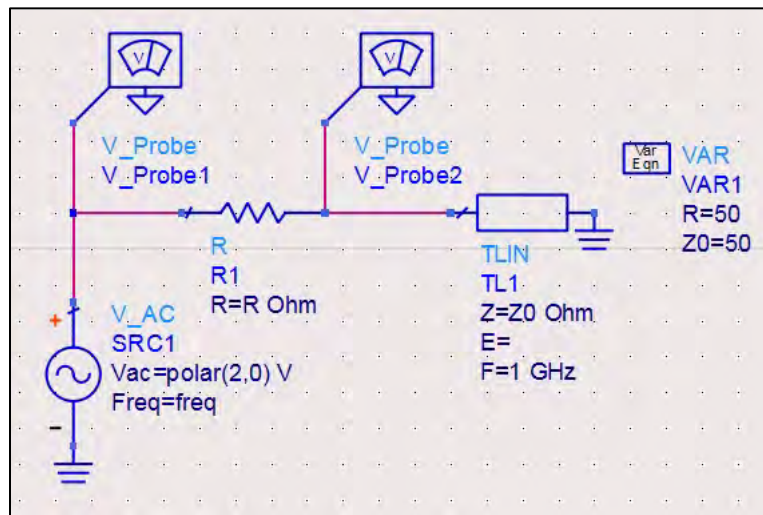


Figure 1-18. Schematic for the short circuit transmission line.

In order to obtain a plot of reactance or voltage versus electrical length, a parameter sweep will be done for the value of theta, in degrees. Create a variable for theta and place it in the electrical length spot for the ideal transmission line. The Parameter Sweep and AC analysis components are found in the Simulation-AC palette. Since the electrical length variation is the goal in this problem, the frequency range for simulation can be a single frequency point. The parameter sweep should reference the variable theta, the simulation plan, and the sweep range of 0 to 360 degrees (Figure 1-19).

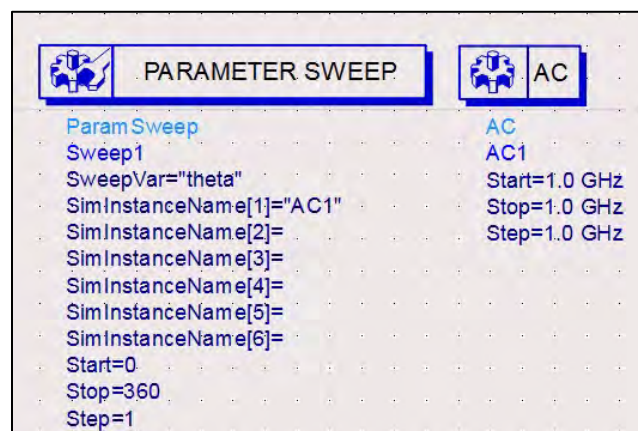


Figure 1-19. The parameter sweep referencing the variable theta, the simulation plan, and the sweep range.

After adding the output variables as measured equations for Z0 and R, the final resulting schematic— ready for simulation—will look like that shown in Figure 1-20.

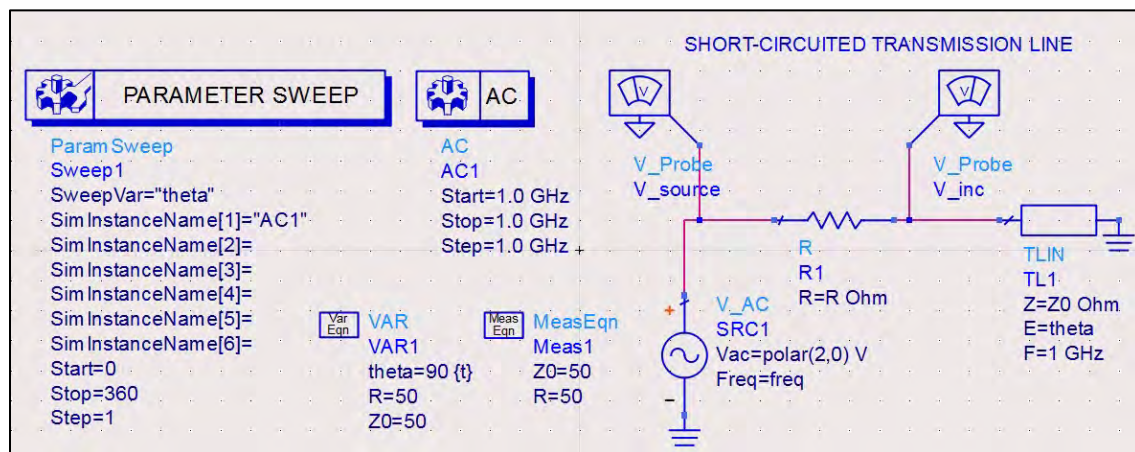


Figure 1-20. The final schematic ready for simulation.

In the Data Display, the following equations are set up to provide the variables necessary to generate the plots for reactance, voltage and current (Figure 1-21). Using probes requires two steps. It is more efficient and cleaner to use named voltage nodes. This chapter will keep the use of probes for a more physical interpretation, but subsequent chapters will name voltage nodes as the skill level of the user progresses.

```
Eqn V_s = V_source.net
Eqn V_in = V_inc.net
Eqn I_in = (V_s-V_in)/R
Eqn Z_in = V_in/I_in
```

Figure 1-21. The following equations provide the variables necessary to generate reactance, voltage and current plots.

The first rectangular graph for reactance is created by plotting the imaginary part of Z\_in with respect to theta. This can be done by manually typing the desired code into the Plot Traces & Attributes window.

Type `plot_vs(imag(Z_in),(theta))` into the text box and click OK (Figure 1-22).

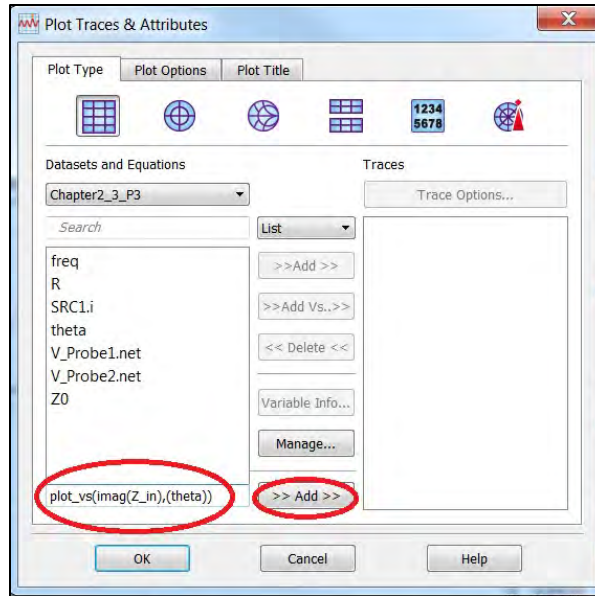


Figure 1-22. Plot Traces & Attributes window.

The plot axes will need to be adjusted for optimal viewing (Figure 1-23). The y-axis can be changed to  $\pm 600$ , and the x-axis from 0 to 360 degrees in 45 degree intervals.

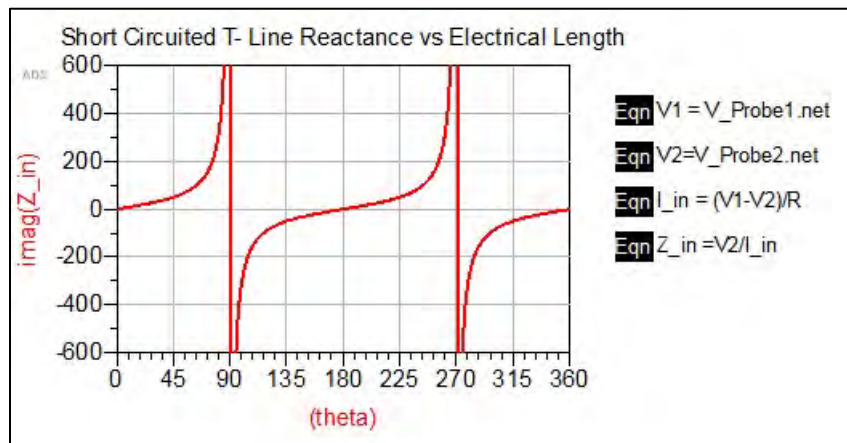


Figure 1-23. Adjustment of the plot axes for optimal viewing.

From the plot, it can be seen that as the electrical length varies, the reactance of the transmission line alternates between inductive and capacitive impedance. While this would be sufficient, it is better to look at the change in terms of lambda,  $\lambda$ . Double click on the trace line and manually change the variable from theta to theta/360 (Figure 1-24).

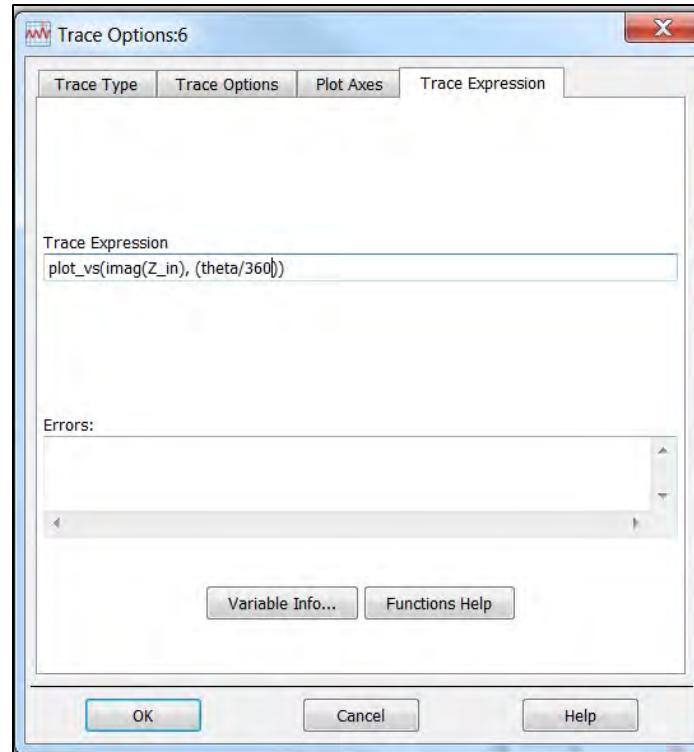


Figure 1-24. Trace Options window.

Making some cosmetic corrections to the axis labels, the reactance versus lambda graph is now complete (Figure 1-25).

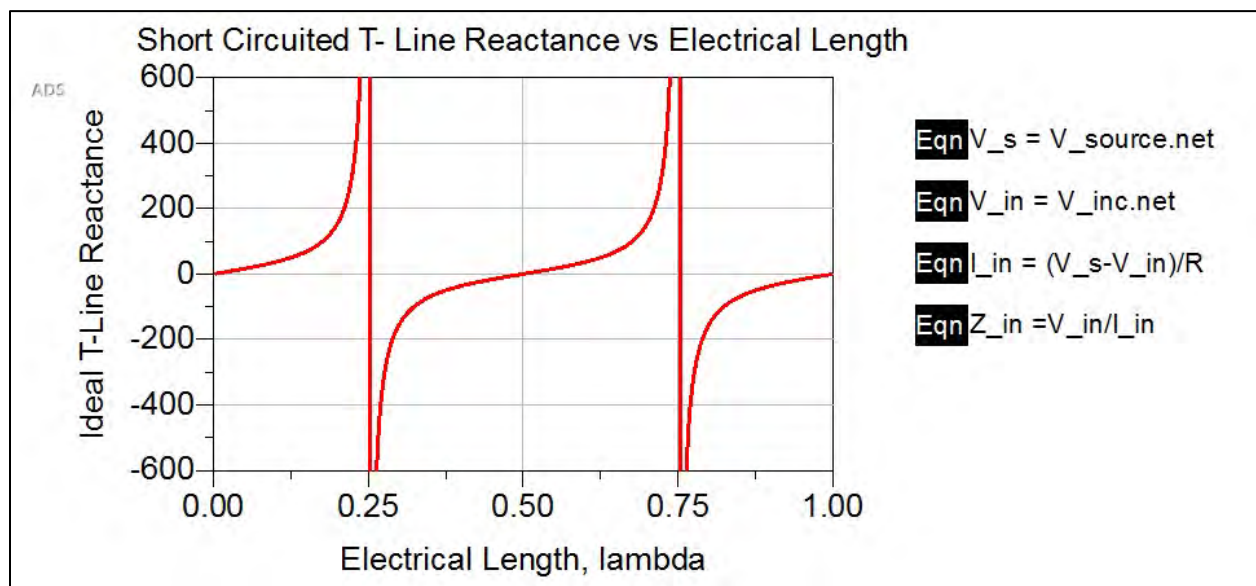


Figure 1-25. The completed reactance versus lambda graph.



The plot for the voltage,  $\text{plot\_vs}(\text{mag}(V_{\text{in}}), (\theta/360))$ , and current,  $\text{plot\_vs}(\text{mag}(I_{\text{in\_mA}}), (\theta/360))$ , are also completed to show their respective changes versus  $\lambda$  (Figure 1-26).

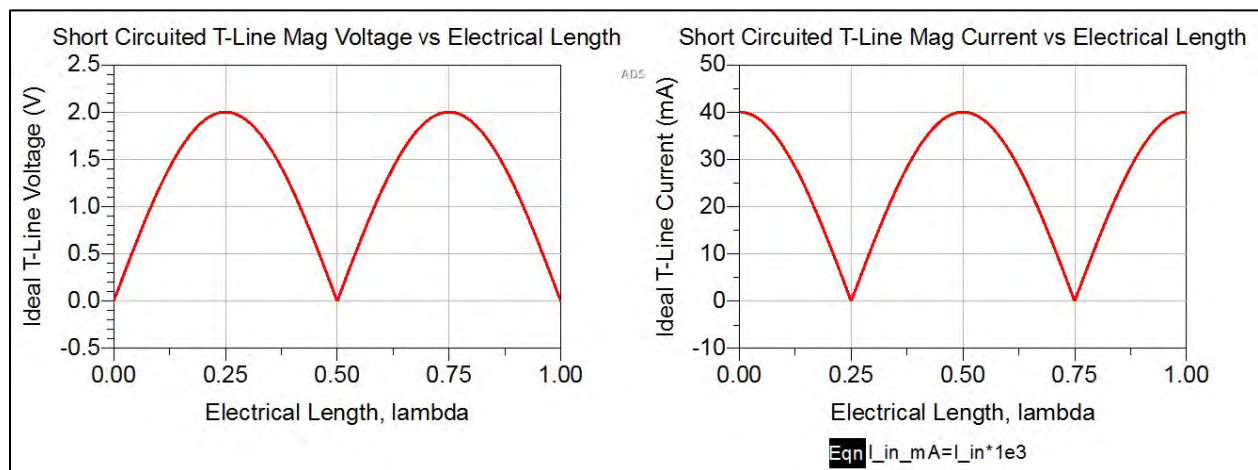


Figure 1-26. Plots for voltage (left) and current (right).

### Problem 3: The Open Circuit Terminated Ideal Transmission Line

#### Problem Statement

Show graphically in ADS, the changes in reactance, magnitude voltage, and magnitude current with respect to line length for an ideal transmission line that is terminated in an open circuit. Compare the results for the open circuit to that of the short circuit.

#### Solution

##### Strategy

Copy the schematic for the short circuit and replace the short termination with a large resistor to simulate an open circuit.

##### What to expect

The impedance mismatch between the 50-Ohm source and open circuit termination will again cause reflections on either side of the transmission line, similar to the short circuit. The characteristic equation for input impedance for a transmission line is given by Equation 1-10:

$$Z_{\text{in}} = Z_o \frac{Z_L + jZ_o \tan \beta l}{Z_o + jZ_L \tan \beta l}$$

Equation 1-10

Equation 1-10 can be reduced to  $Z_{in} = -jZ_o \cot \beta l$ , for the open circuit case. When the line length is  $0 \lambda$  in length, we expect to see the open circuit termination only,  $Z_{in} = \infty$ . The voltage and current will measure 2 V and 0 mA, and the reactance will be  $\infty$  Ohms. When the line length is a  $0.25 \lambda$ , the input impedance will look like a short, making the reactance zero, the voltage 0 V and current 40 mA. When the line length is  $0.5 \lambda$ , the transmission line will look like an open circuit again. There will be a phase change, but because we are plotting the magnitude voltage and current, the values will be the same as the  $0-\lambda$  line length.

## Execution

Following the same procedure as Problem 2, the final schematic for the open circuit terminated ideal transmission line is given in Figure 1-27.

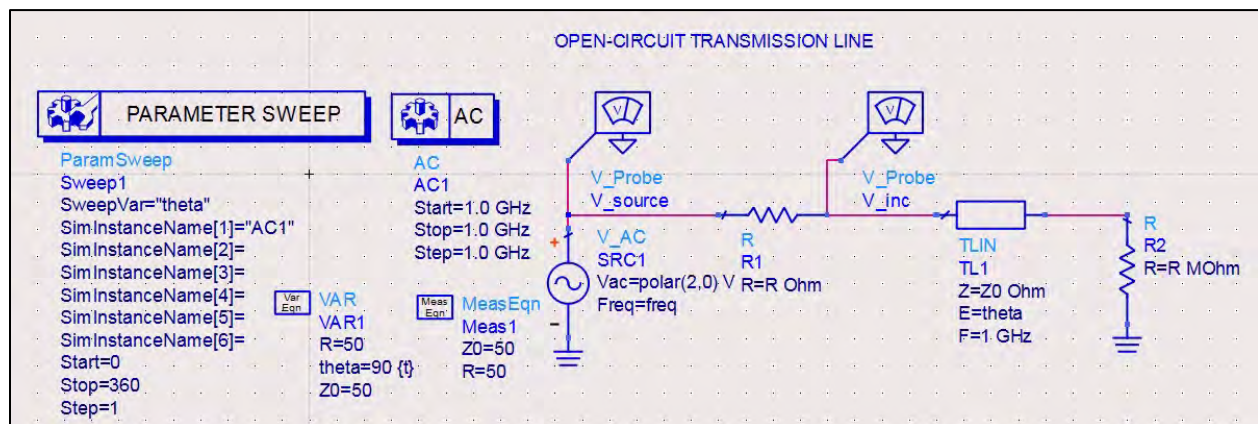


Figure 1-27. Open circuit, terminated ideal transmission line schematic.

In the Data Display window, the following plots for the open circuit reactance, voltage and current are plotted against lambda, similar to Problem 2 (Figure 1-28).

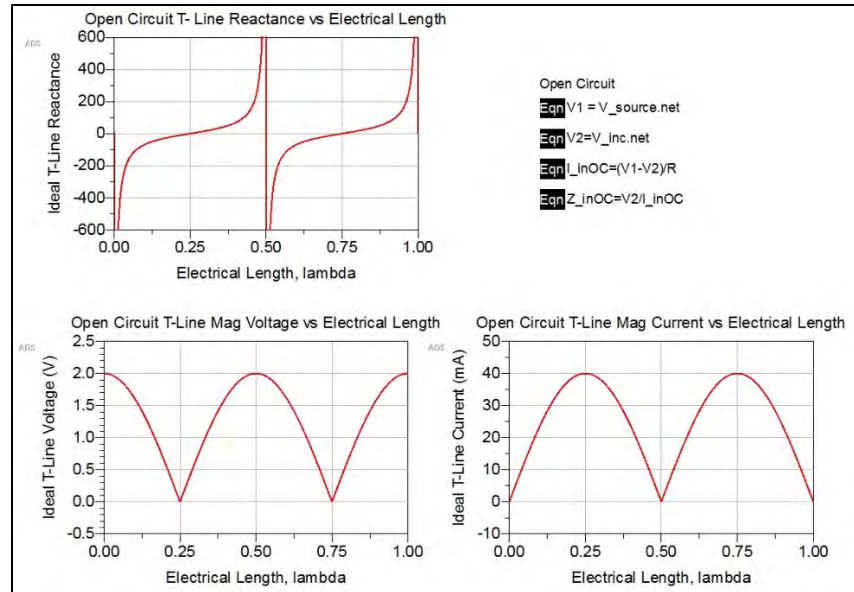


Figure 1-28. Shown here are the plots for the open circuit reactance (top), voltage (bottom left) and current (bottom right) plotted against lambda.

In order to graphically compare the short circuit and open circuit simulations, a second set of equations are created for the short circuit parameters. To grab data from a different simulation and plot it in the current simulation, open the Enter Equation window (Figure 1-29). The drop down menu on the top right is the simulated data selector. Select the Problem 2 data set, then the specific variable, and insert it into the equation text field.

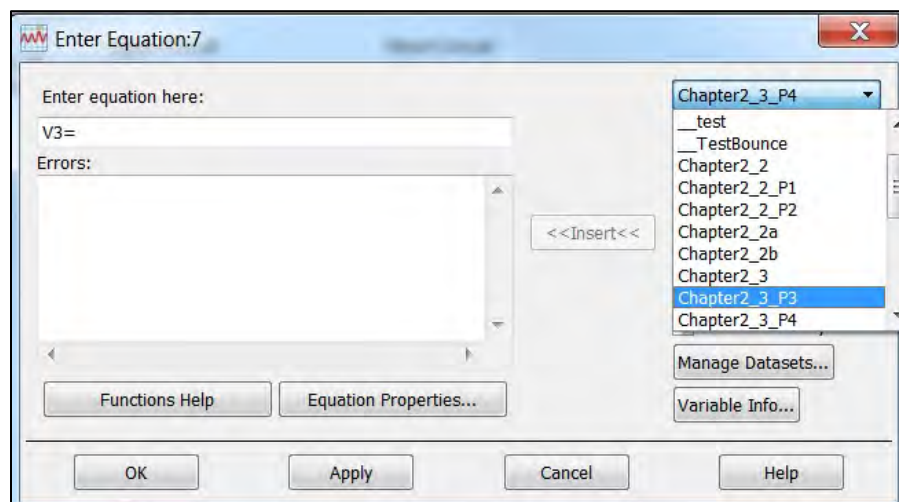


Figure 1-29. Enter Equation window.

The resulting set of equations are created for both the open and short circuit simulations (Figure 1-30). Even though the value of the generator impedance,  $R$ , is the same for both simulations, it is



better to keep the data items separate and not reuse the R value for the open circuit simulation in the short circuit input current equation.

| Open Circuit                 | Short Circuit                              |
|------------------------------|--------------------------------------------|
| <b>Eqn</b> V1 = V_source.net | <b>Eqn</b> V3=Chapter2_3_P3..V_source.net  |
| <b>Eqn</b> V2=V_inc.net      | <b>Eqn</b> V4 = Chapter2_3_P3..V_inc.net   |
| <b>Eqn</b> I_inOC=(V1-V2)/R  | <b>Eqn</b> I_inSC=(V3-V4)/Chapter2_3_P3..R |
| <b>Eqn</b> Z_inOC=V2/I_inOC  | <b>Eqn</b> Z_inSC=V4/I_inSC                |

Figure 1-30. Resulting equations for the open and short circuit simulations.

To plot both reactance traces on the same plot, simply type both equations into the Plot Traces & Attributes window (Figure 1-31).

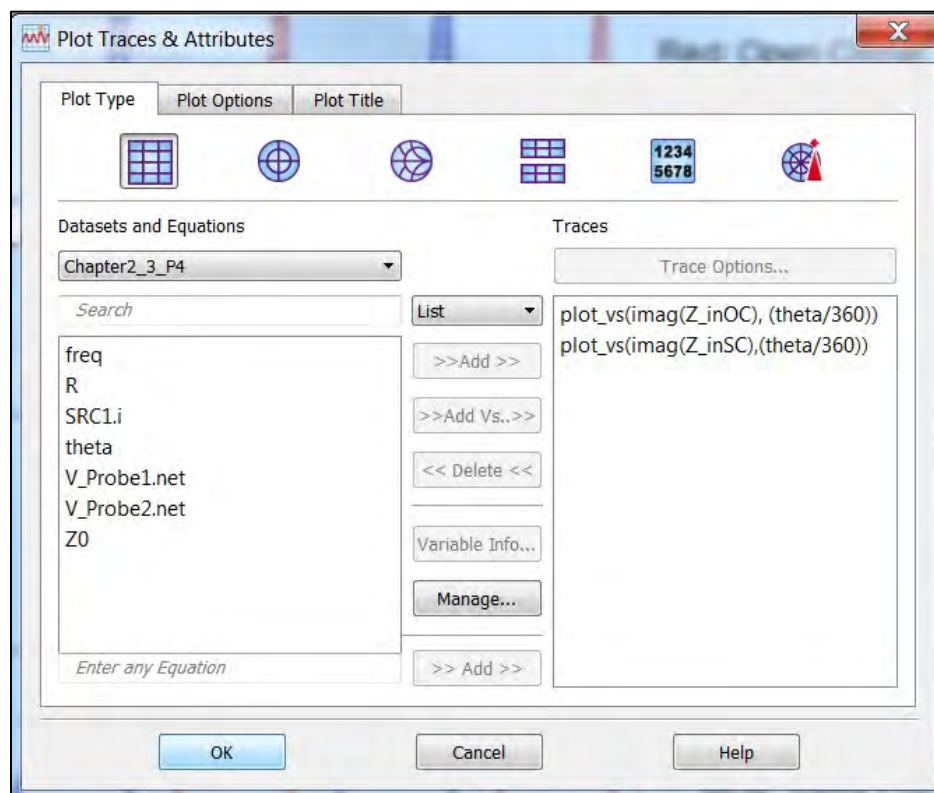


Figure 1-31. Plot Traces & Attributes window.

The three final comparison plots are provided in Figure 1-32.

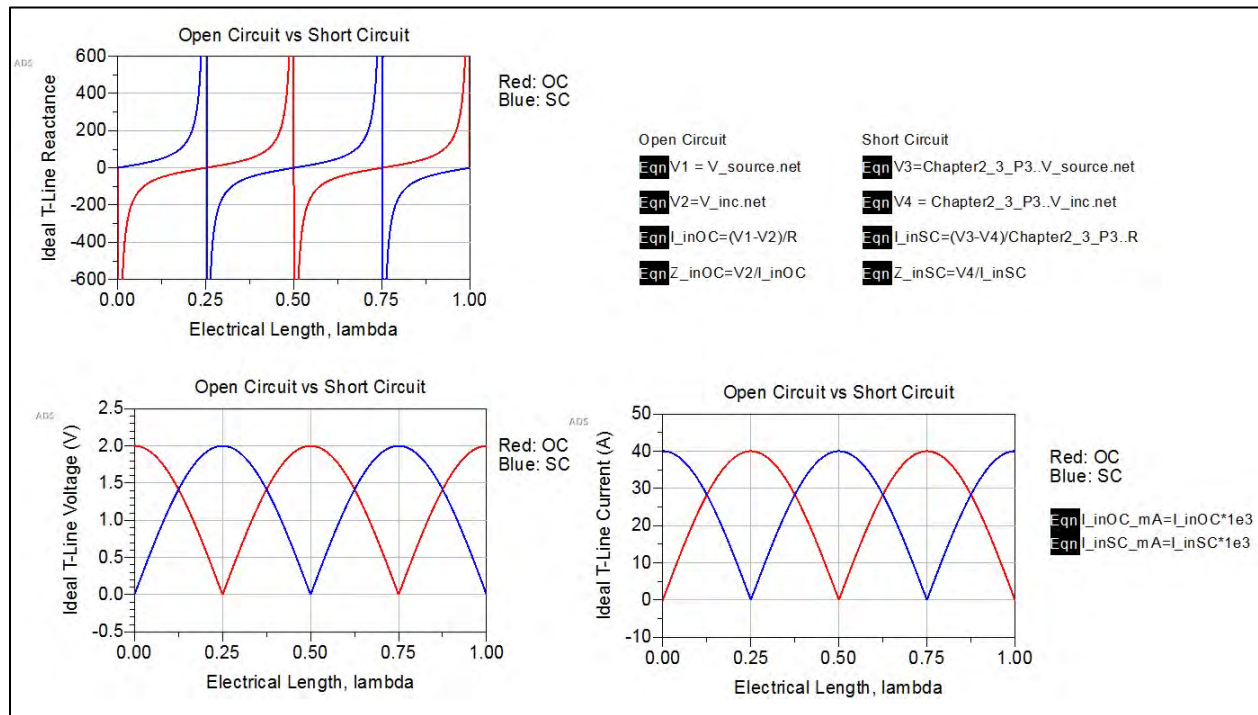


Figure 1-32. Open circuit versus short circuit comparison plots.